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Air Sensor Network Analysis Tool: R-Shiny application

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Summary

An R-Shiny application has been developed to assist air quality professionals in 1) comparing air sensor performance to nearby ambient air reference monitors, 2) applying basic quality assurance (QA) and quality control (QC), and 3) understanding local air quality conditions.

Statement of need

Poor air quality is associated with negative health effects (Manisalidis et al. 2020). Air quality data are needed to understand local conditions and reduce exposure to air pollution. In the U.S., air quality is measured by ambient air monitors operated by state, local, and Tribal air agencies (*Ambient Air Quality Surveillance*, 2006). Recent efforts supplement the national monitoring network with localized air sensor data to investigate variations at neighborhood scales. Air sensors typically cost one or two orders of magnitude less than conventional air monitors (e.g., $10-$10k). However, air sensor data may be noisy (Barkjohn et al., 2025; Zoest et al., 2018), biased, or inaccurate (Karagulian et al., 2019). Furthermore, sensor performance may vary over time (deSouza et al., 2023), concentration range, or the environment in which they operate (e.g., high RH) (Giordano et al., 2021; Levy Zamora et al., 2022). Sensors often must be compared with nearby measurements and corrected to improve the data quality (Barkjohn et al., 2022; Collier-Oxandale et al., 2020; Feenstra et al., 2019; Zheng et al., 2018). The air sensor data analysis required to account for these limitations is challenging for users without extensive coding experience and for agencies faced with increased data volumes and community questions.

Many existing air data analysis software tools require coding experience to be run in an open-source environment (Carslaw & Ropkins, 2012; Collier-Oxandale et al., 2022); are specific to one manufacturer's sensor (Collier-Oxandale et al., 2022; Feenstra et al., 2020); or are designed to focus on single collocation sites (Frederick & Kumar, 2024). An analysis tool is needed to more easily aggregate data from multiple air quality data sources, perform standard QC, identify nearby sensors spatially, and create comprehensive visualizations. Generating air quality analysis in this form will further the capabilities of air sensors and potentially aid in protecting public health.

Overview

The Air Sensor Network Analysis Tool (ASNAT) is an R-Shiny (Chang et al., 2025; R Core Team, 2024) application that integrates data from multiple sensor networks to provide a

broader representation of local air quality. This tool can perform data correction as well as QC flagging to improve data quality so users can better understand the performance of new air sensor networks (including during extreme events like wildfires, dust storms, and fireworks). Air quality professionals who have base knowledge of air quality analysis are the target users of ASNAT.

Users can select data from EPA's Remote Sensing Information Gateway (RSIG) including AirNow, EPA's Air Quality System (AQS), Meteorological Aerodrome Report (METAR), and PurpleAir (Szykman, Plessel, and Freeman 2012) (https://www.epa.gov/hesc/web-access-rsig-data, last accessed 2/28/25) (Figure 1). Data from AirNow and AQS available in ASNAT include fine particulate matter ($PM_{2.5}$), particulate matter 10 µm or less in diameter (PM_{10}), relative humidity (RH), temperature, pressure, ozone (O_3), nitrogen dioxide (NO_2), carbon monoxide (CO), and sulfur dioxide (SO_2). The PurpleAir dataset includes the corrected PurpleAir $PM_{2.5}$ data (PurpleAir.pm25_corrected) data that is corrected in the same way as PurpleAir data on the EPA and US Forest Service Fire and Smoke Map (fire.airnow.gov, accessed: 1/28/25) (Johnson Barkjohn et al. 2021). Users can also load data from standard format text files, which can be generated using the Air Sensor Data Unifier tool (https://www.epa.gov/air-sensor-toolbox/air-sensor-data-tools, last accessed 2/28/25), or from offline PurpleAir files.

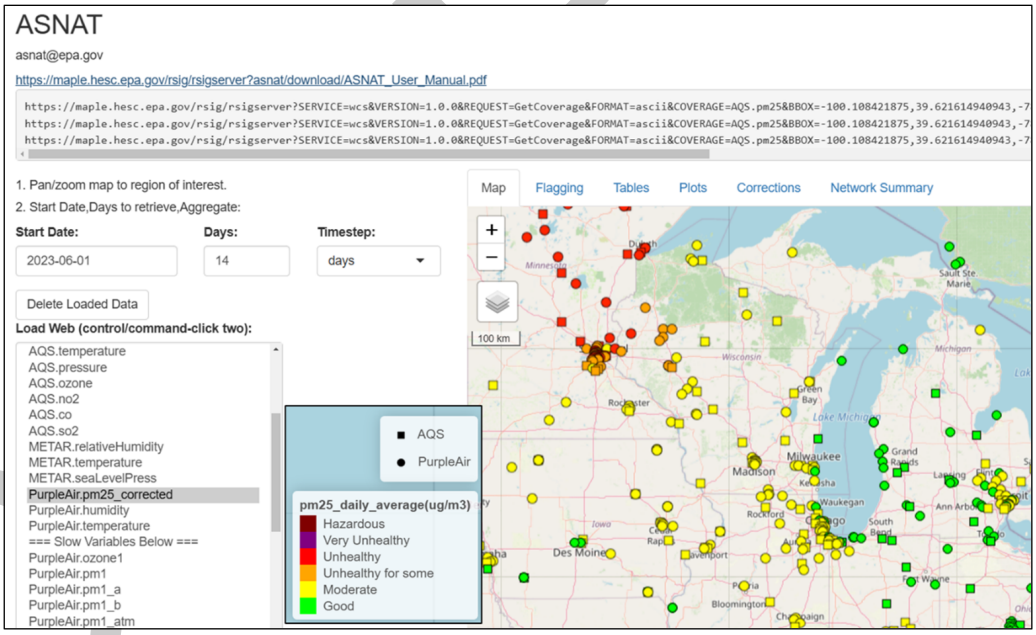

Figure 1: Figure 1. Screenshot of ASNAT showing data selection on the left and loaded data on the right. The display includes AirNow and corrected PurpleAir sensor data across the Midwestern US and has multiple tabs to further quality assure and visualize the data.

ASNAT provides a variety of ways for users to understand air quality. On the home page, users can step through the hourly or daily measurements on a geospatial representation (Figure 1). The plots and network summary tabs include a variety of visualizations to understand network performance and local air quality (Figure 2).

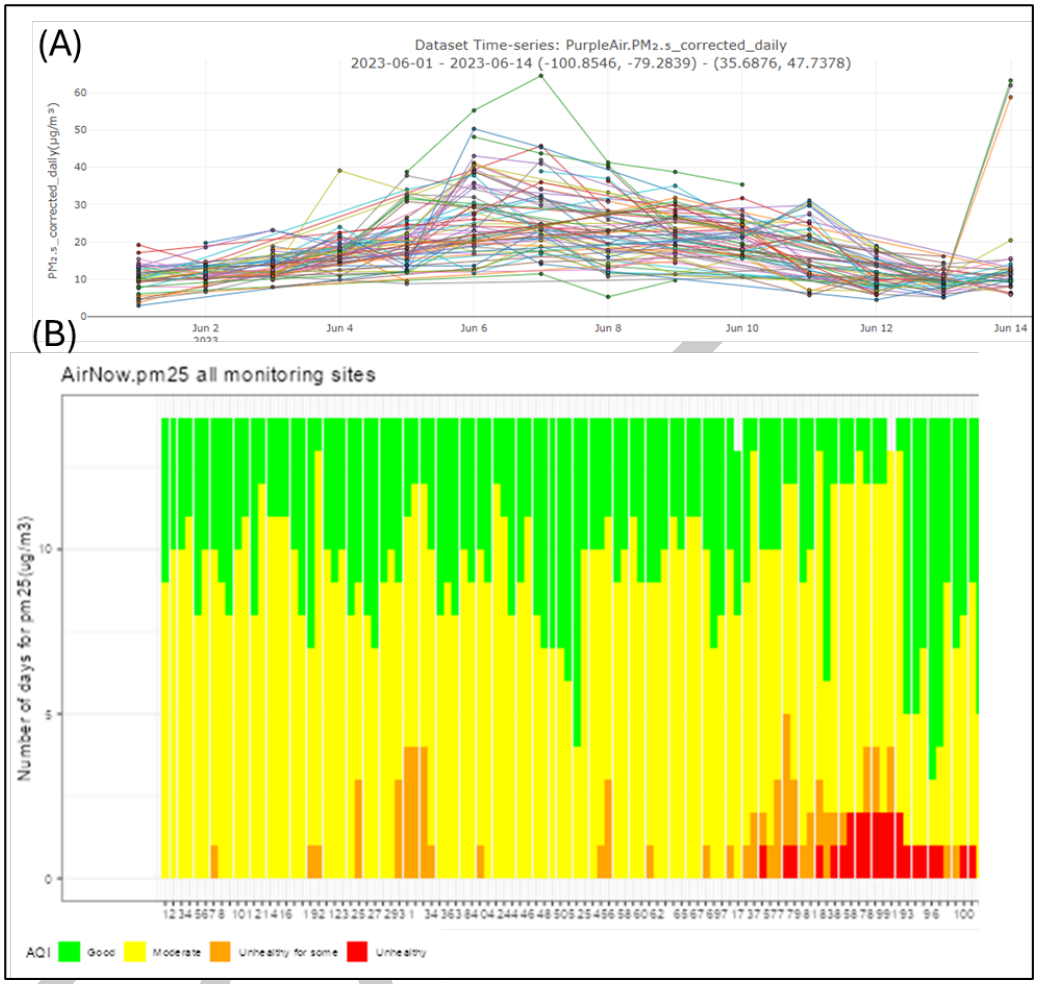

Figure 2: Figure 2. A) Shows the daily concentrations with each line indicating a different PurpleAir sensor. B) Shows the number of days in each AQI category grouped by site ID.

ASNAT allows users to compare two datasets by finding nearest neighbors within a user specified distance. Typically, this involves comparing a sensor dataset with more uncertainty and unknown performance to reference monitoring data (e.g., AirNow, AQS) to better understand sensor performance and develop corrections (Figure 3). ASNAT provides similar performance metrics as outlined in EPA's performance targets (Duvall et al., 2021b, 2021, 2023, 2024).

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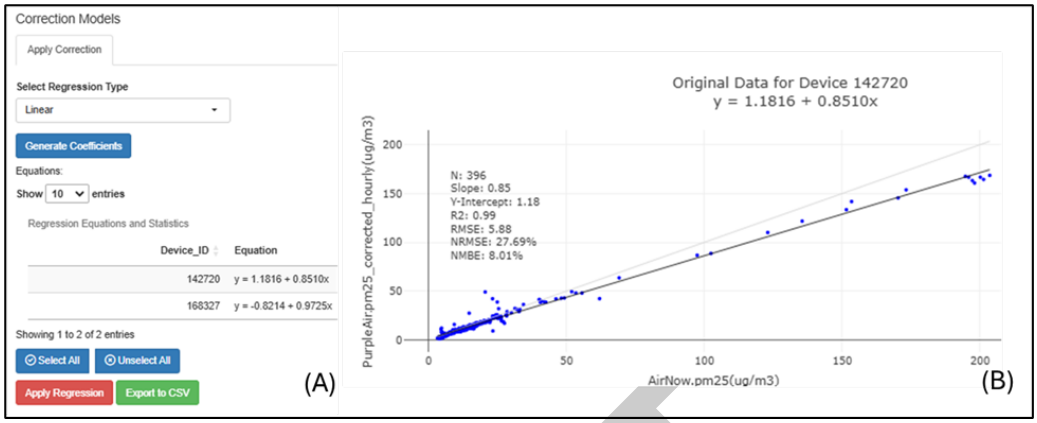

Figure 3: Figure 3. (A) Applying a linear regression to two sensor monitor pairs and (B) a scatter plot showing performance statistics for a PurpleAir AirNow pair.

This tool provides a variety of options to flag and/or remove anomalous data based on several QA categories such as: exceedance of a threshold, agreement with a nearest neighbor, repeated values, outlier detections (using several methods), a user specified period (e.g., a day with a known sensor issue), and more (Figure 4).

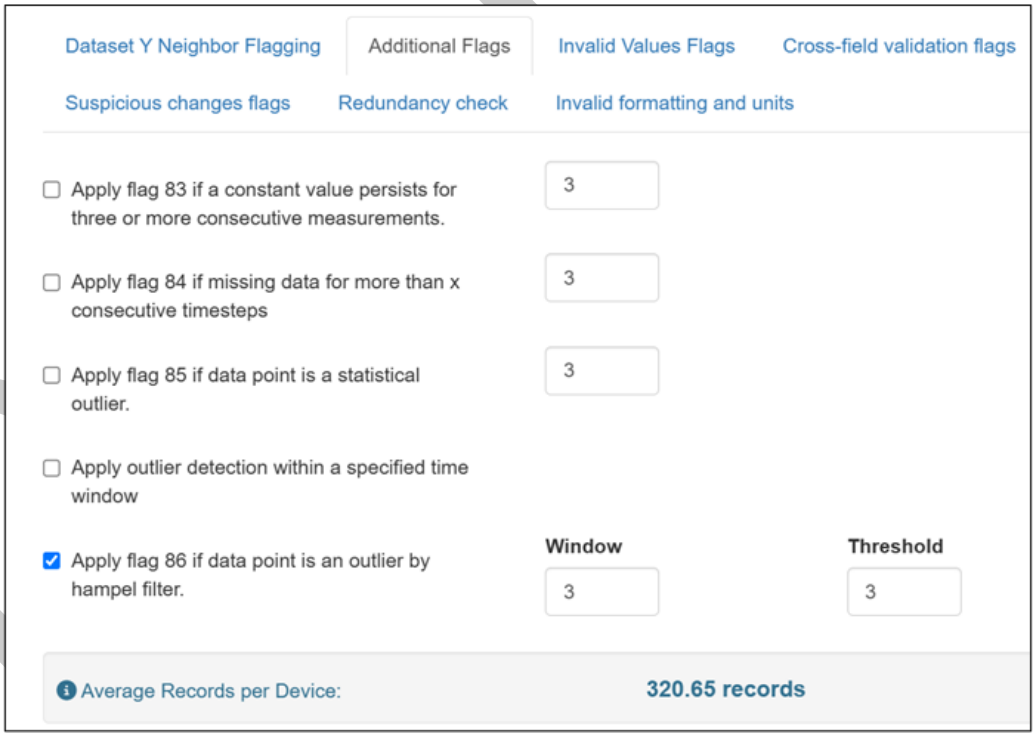

Figure 4: Figure 4. ASNAT flagging functionality.

Limitations

- Dataset size limits are dependent on local computing resources. Users should limit longer time analysis to smaller spatial ranges and/or longer averaging intervals.
- Some background air quality knowledge is needed to successfully use this tool and make informed decisions based on ASNAT outputs. For example, nearest neighbor radius will depend

on pollutant chemistry, sources, and geography. Local knowledge may be required to determine if outliers are due to sensor malfunctions or real short-term pollutant events.

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